

Paolo Cremonese, Ph.D.

Researcher

Vancouver, BC

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PaoloCremo

About me

Data Scientist with a Postdoctoral research background in gravitational waves and a strong foundation in data analysis, machine learning, and AI. Passionate about fintech, technology, and continuous learning. Based in Vancouver, actively exploring the next career move.

Skills

Python - pandas, numpy, matplotlib, tensorflow, scikit-learn

SQL

Linux

LaTeX

Git

HTML/CSS

Mathematica

Slurm - HTCondor

Languages

Italian - Mother Tongue

English - Fluent

Spanish - Basics

Interests

Technology

AI/ML

Data Analysis

Finance and Investments

Sports, Books & Piano.

Work Experience

2026-now **Physics SME** — Mercor, Remote

Contract role contributing to AI model improvement. Evaluated AI-generated physics content for technical accuracy, provided domain-specific corrections, and refined evaluation criteria and quality standards.

2022-2024 **PostDoc Researcher** — UIB, Palma, Spain

Data Scientist. Member of LIGO collaboration.
(i) I **developed models to analyze Gravitational Waves data**.(ii) I co-led a project to **automate the process of studying lensed events** more efficiently. (iii) I coded **a pipeline to automate the creation of setup files**, for cluster computing, for bayesian parameter estimation (PaoloCremo/runtomation).

2018 **Internship as Data Scientist** — DBA Group, Italy

I worked on developing methods to record and analyse diverse datasets, from meteorological to industrial data, **using machine learning techniques to predict outcomes and improve data-driven decision-making**

Education

2022 **Ph.D. in Physics** — Szczecin, Poland

Gravitational lensing of Gravitational Waves

2017 **M.Sc. in Astronomy** — Padova, Italy & Stockholm, Sweden

Thesis on lensing of gravitational waves done at Stockholm university

2015 **B.Sc. in Astronomy** — Padova, Italy

Thesis on dark matter in spiral galaxies

Projects

myFinances — iOS app (SwiftUI, C++, SQL) to track personal finances.
Travel Buddy — full-stack AI app using Agentic LLMs, RAG, and prompt engineering.
Paperbot — Python bot for daily ArXiv paper updates via Telegram.
paolocremonese.com — personal website, self-coded and maintained.

Volunteering

2025 **Partnerships Lead** — Data For Good, Vancouver, Canada

I engage nonprofit organizations to identify their data needs, develop tailored strategies for impactful projects, and implement a CRM system to streamline client outreach and relationship management.

Main Publications

#6 **Invariance transformations in wave-optics lensing: implications for gravitational-wave astrophysics and cosmology**
#5 **Follow-up Analyses to the O3 LIGO-Virgo-KAGRA Lensing Searches**
#4 **Characteristic features of Gravitational Wave lensing as probe of lens mass model**
#3 **Breaking the mass-sheet degeneracy with gravitational wave interference in lensed events**

35+ papers, 1300+ citations. Full list: inspirehep.net | Google Scholar

Conferences Presentations

Presented my research on gravitational wave lensing at 10+ international conferences and seminars (2019–2023), including LVK meetings, GWverse, COSMO '21, and Amaldi 14. Also gave a public divulgative talk on dark matter (YouTube, 2022).